

From Circuits to Claims: Post-Selection Certificates for Mechanistic Interpretability

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Abstract

Circuit discovery in mechanistic interpretability increasingly relies on adaptive selectors, counterfactual datasets, and standardized evaluations. We study selected circuits as post-selection objects and define a mechanistic claim by its model, computational graph, task distribution, intervention design, task score, selector, selection data, selected circuit, and audited properties. We introduce Circuit Certificates, a claim-centered audit framework reporting behavioral preservation, localization, edgewise minimality, minimality profiles, selector stability, intervention robustness, power, and structural recovery when ground truth is available.

The framework implements fixed holdout, folded fixed-circuit audit, selective split, and selector-level cross-fit protocols. Selective split yields finite-sample post-selection validity for any measurable selector under independent audit data. Selector-level cross-fit audits the selection procedure through a procedure-level risk estimator. Edgewise minimality is assessed with Holm-adjusted tests, with minimality profiles summarizing the distribution of edge effects.

We instantiate the framework in an audited artifact covering 65 run packets and 158 certificates. The results identify distinct evidence regimes across tasks. On GPT-2 small IOI, selector-level cross-fit certifies localization with high selector stability and adequate power, while preservation and minimality are uncertified. On Gemma2 ARC-Easy, it certifies preservation while localization is uncertified. On Gemma2 arithmetic subtraction, it certifies localization in a lower-power regime. On InterpBench, behavioral support coincides with weak edge-level structural recovery. These findings support reporting mechanistic interpretability results as fully specified audited claims.

1 Introduction

Circuit analysis has become a central mode of mechanistic interpretability. Language-model circuits are discovered with activation patching, path patching, attribution-based surrogates, and mask-optimization methods. Canonical tasks

such as indirect object identification and arithmetic now have established datasets and counterfactual constructions. Benchmark suites increasingly standardize tasks, metrics, and selectors [15, 4, 1, 6, 14, 5, 12]. This infrastructure makes circuit discovery more systematic and sharpens the evidential question attached to any returned subgraph: which mechanistic claim does it support?

A selected circuit is the output of an adaptive pipeline whose construction depends on graph granularity, task distribution, counterfactual construction, scoring rule, selector class, calibration rule, and often a size threshold. The returned subgraph therefore inherits the assumptions and degrees of freedom introduced by those choices, while its evidential status depends on the audit protocol, the relation between audit and selection data, the intervention family used for evaluation, the multiplicity induced by edgewise testing, the selector’s stability under resampling, and, when available, the relation between behavioral support and structural recovery. Recent work identifies these axes as sources of inferential variation: intervention design changes the estimand of faithfulness metrics [11, 7, 8]; candidate-circuit evaluation admits a formal testing treatment [14]; circuit discovery behaves as a statistical estimation pipeline with substantial instability [10]; and benchmarks separate causal localization from structural recovery [5, 12]. A reporting standard for circuit discovery should specify the claim, the selection procedure, the audit target, and the uncertainty attached to the result.

The framework introduced here takes the *mechanistic claim* as its central object: a selected circuit together with its model, graph abstraction, task distribution, intervention design, score, selector, selection data, and audited property. Specifying claims at this level gives post-selection inference a direct role in circuit auditing. The resulting framework, which we call *Circuit Certificates*, addresses the following research question:

Given a discovered circuit, which claim does the audit support, under which intervention design and protocol, and with what statistical uncertainty?

Formalization. We define a mechanistic claim as an inferential object indexed by a model, graph, task distribution, intervention design, task score, selector, selection data, and selected circuit. This vocabulary assigns separate audit targets to behavioral preservation, localization, edgewise minimality, selector stability, intervention robustness, power, and structural recovery.

Theory. We develop four audit protocols: fixed holdout, folded fixed-circuit audit, selective split, and true selector-level cross-fit. The audit target may be a fixed circuit or the selector that produces circuits. For selective split, we prove finite-sample validity under independent audit data for any measurable selector. For true selector-level cross-fit, we identify the target as a procedure-level out-of-fold estimand and separate foldwise validity from stronger aggregate concentration claims. We also obtain familywise-valid edgewise minimality decisions through Holm adjustment.

Evidence. We instantiate the framework in a protocol-audited artifact and evaluate it on canonical and benchmarked settings. The empirical results show repeated separations between evidence dimensions that are often compressed into a single circuit-quality score.

The audits yield distinct claim profiles across regimes. On GPT-2 small IOI, true selector-level cross-fit supports localization with high selector stability and adequate power; preservation, minimality, and strong robustness remain unsupported. On Gemma2 ARC-Easy, the audit supports preservation with adequate power and leaves localization unsupported. On Gemma2 arithmetic subtraction, a second non-IOI case exhibits selector-level localization in a weaker, low-resolution regime. On InterpBench, selector-level behavioral support co-exists with weak edge-level structural recovery, separating behavioral support from ground-truth circuit recovery. A final MCQA control shows that selector consistency contributes one layer of evidence within a broader certificate. Together, these results support a reporting practice in which discovered circuits are presented through audited claim profiles.

The framework builds on the circuit perspective of Olah et al. [13], Elhage et al. [2]; the intervention-aware methodology of path patching and activation patching [4, 7]; the semantic foundations supplied by causal abstraction [3]; fixed-circuit testing in Shi et al. [14]; and the empirical anchors supplied by InterpBench and MIB [5, 12]. The result is a rigorous and executable framework for moving from discovered circuits to auditable mechanistic claims.

2 Mechanistic claims and audit protocols

2.1 Claims

Let M be a model represented at a fixed graph granularity by a directed acyclic graph $\mathcal{G} = (V, \mathcal{E})$. Let $\mathcal{D}$ denote a task distribution over tuples $(x, \tilde{x}, y)$, where x is a clean input, $\tilde{x}$ is a counterfactual input used for intervention, and y is task-level ground-truth information when required by the score. Let s be a scalar task score, and let η denote an intervention design, such as complement patching with counterfactual activations, mean ablation, or optimal ablation. A selector $\mathcal{A}$ maps selector-visible data to a circuit $C \subseteq \mathcal{E}$.

Definition 1 (Mechanistic claim). *A mechanistic claim is a tuple*

$$\Gamma = (M, \mathcal{G}, \mathcal{D}, \eta, s, \mathcal{A}, S, C, \mathcal{P}),$$

where S is the selector-visible data, $C = \mathcal{A}(S)$ is the selected circuit, and $\mathcal{P}$ is a family of audited properties.

The tuple records the information needed to interpret the evidence attached to the selected circuit, including the model, graph abstraction, task distribution, intervention design, score, selector, selection data, and audited properties. Because graph choices, counterfactual constructions, scores, and intervention families determine what the selected subgraph is evidence for, the audit treats the claim as the evidential unit.

2.2 Behavioral properties

We audit two behavioral properties that are central to circuit evaluation. Let $M_C^\eta(x, \tilde{x})$ denote the output of the selected circuit under intervention design η , and let $M_{\bar{C}}^\eta(x, \tilde{x})$ denote the output of the complement construction under the same design. Let ℓ be a bounded discrepancy between scores, logits, or task outputs.

Behavioral preservation. We define

$$R_{\text{pres}}^\eta(C) = \mathbb{E}_{(x, \tilde{x}, y) \sim \mathcal{D}} [\ell(M(x), M_C^\eta(x, \tilde{x}))].$$

Small values indicate that the selected circuit preserves the behavior of the full model under the chosen intervention design. This quantity measures behavioral sufficiency for the selected circuit.

Localization. We define

$$R_{\text{loc}}^\eta(C) = \mathbb{E}_{(x, \tilde{x}, y) \sim \mathcal{D}} [\ell(M(x), M_{\bar{C}}^\eta(x, \tilde{x}))].$$

Large values indicate that the complement construction loses the target behavior, so the behavior is concentrated in the selected circuit under the chosen intervention design.

Preservation quantifies behavioral sufficiency. Localization quantifies behavioral concentration. A circuit may preserve the target behavior while its complement also preserves it; another may concentrate the target behavior while failing to reproduce the full model on its own. The experiments below exhibit both regimes.

Edgewise minimality. For each edge $e \in C$, define the edge effect

$$\Delta_e(C; \eta) = R_{\text{pres}}^\eta(C \setminus \{e\}) - R_{\text{pres}}^\eta(C).$$

This quantity measures the loss in behavioral preservation produced by removing edge e from the selected circuit. Large circuits induce a family of edgewise tests, so minimality auditing requires multiplicity control.

Minimality profiles. A scalar minimality decision gives a coarse summary of edgewise structure. We report the profile

$$\text{MinProf}_C(\epsilon) = \frac{1}{|C|} \sum_{e \in C} \mathbf{1}[\Delta_e(C; \eta) > \epsilon].$$

As ϵ increases, $\text{MinProf}_C(\epsilon)$ records the fraction of edges whose effect exceeds the threshold. The profile distinguishes circuits with a descriptive high-effect tail from circuits with a broader low-effect fringe.

Selector stability. Given bootstrap or foldwise selector outputs $C^{(1)}, \dots, C^{(B)}$, stability is summarized by inclusion probabilities $\pi_e = \frac{1}{B} \sum_b \mathbf{1}[e \in C^{(b)}]$, mean pairwise Jaccard overlap, selection entropy, and thresholded core/fringe edge sets. These summaries provide a circuit-valued analogue of stability selection [9].

Intervention robustness. Intervention design changes the estimand [11, 7, 8], so claims are indexed by an intervention family $\mathcal{H}$. A certificate reports preservation and localization under each member of $\mathcal{H}$, together with preservation robustness, localization robustness, strong robustness, and a sensitivity norm.

Structural recovery. When a benchmark provides a ground-truth circuit C^* , the selected circuit can also be evaluated structurally through precision, recall, and F1. Structural recovery is reported alongside behavioral auditing, since edge overlap and behavioral support measure distinct properties of the selected circuit.

2.3 Audit protocols

Each claim includes an audit protocol, chosen from four cases:

1. **Fixed holdout:** a fixed externally supplied circuit is audited on independent holdout data.
2. **Folded fixed-circuit audit:** a fixed circuit is audited over multiple folds, with selection performed once.
3. **Selective split:** discovery and calibration are performed on selector-visible data, and the selected circuit is audited on an independent audit split.
4. **True selector-level cross-fit:** for each fold, the circuit is rediscovered without using that fold and audited on the held-out fold.

Selective split supports a selected-circuit claim for the circuit returned by one adaptive pipeline. True selector-level cross-fit audits the selection procedure by repeatedly rediscovering circuits and auditing them on held-out folds. The certificate records the protocol so that each reported property carries the corresponding evidential scope.

3 Circuit Certificates

A *Circuit Certificate* is the audited evidence profile attached to a mechanistic claim. We use *certificate* in the statistical-audit sense: a structured, protocol-indexed record of the statements supported by the audit for the specified claim. The certificate records point estimates, one-sided confidence bounds, protocol metadata, minimality outputs, stability summaries, robustness vectors, structural-alignment results when available, and power calculations.

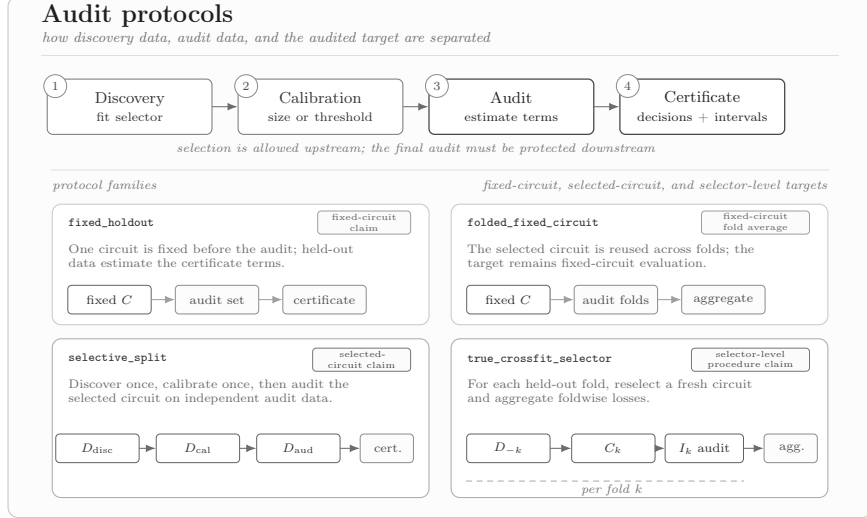


Figure 1: Audit protocols.

3.1 Selective-split validity

Selective split is the basic post-selection protocol. Discovery and calibration may be adaptive; the audit data are held out from both stages. This separation yields finite-sample one-sided certification for any measurable selector.

Theorem 2 (Selective-split post-selection validity). *Let $S = (D_{\text{disc}}, D_{\text{cal}})$ be selector-visible data and let $C = g(S)$ be any measurable selector output. Let D_{aud} be independent of S . For any bounded loss $w(C, Z) \in [a, b]$, define the audit risk*

$$R(C) = \mathbb{E}_{Z \sim \mathcal{D}}[w(C, Z)].$$

Conditional on S , the audit losses are i.i.d. bounded observations with mean $R(C)$. Therefore any valid one-sided concentration bound for bounded i.i.d. means yields a finite-sample valid confidence bound for the selected circuit. In particular, empirical-Bernstein upper and lower confidence bounds are valid for preservation and localization, respectively.

The theorem applies to arbitrary measurable selection pipelines, including manual, attribution-based, graph-pruning, and calibrated threshold procedures. Its identifying condition is independence between the audit data and the data used for discovery and calibration.

3.2 Selector-level cross-fit

True selector-level cross-fit audits the selector as a procedure. For folds $\mathcal{I}_1, \dots, \mathcal{I}_K$, let C_k denote the circuit selected without using fold k , and let $w(C_k, Z_i)$ be the

corresponding audit loss for $i \in \mathcal{I}_k$. The procedure-level target is

$$R_{\text{cf}} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \mathbb{E}[w(C_k, Z_i) \mid C_k].$$

The cross-fit estimator is

$$\hat{R}_{\text{cf}} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} w(C_k, Z_i).$$

Proposition 3 (Foldwise out-of-fold validity). *For each fold k , conditional on the foldwise selected circuit C_k , the held-out losses $\{w(C_k, Z_i) : i \in \mathcal{I}_k\}$ are i.i.d. bounded variables. Consequently, one-sided empirical-Bernstein bounds can be applied validly on each held-out fold for its corresponding foldwise risk.*

Fold-resampled auditing provides selector-level evidence by estimating the behavior of the selection procedure under repeated foldwise discovery and held-out auditing. For the fully aggregated selector-level estimator, an unconditional finite-sample concentration result requires additional dependence-aware assumptions, since data from one audit fold may contribute to the selection of a circuit audited on another fold. The results in Section 5 use this distinction to separate fixed-circuit claims from procedure-level claims.

3.3 Edgewise minimality and multiplicity

Minimality is audited through edgewise hypotheses

$$H_{0,e} : \Delta_e(C; \eta) \leq \tau_{\min}, \quad H_{1,e} : \Delta_e(C; \eta) > \tau_{\min}, \quad e \in C.$$

The framework reports both Holm-adjusted and Benjamini–Hochberg summaries. The formal guarantee is familywise error control.

Proposition 4 (Familywise-valid minimality decisions). *Under selective split or true selector-level cross-fit, if each edgewise p -value is valid conditional on the selected circuit and the audit sample used for that edge, then Holm adjustment controls the familywise error rate at level α for the family of edgewise minimality claims.*

A familywise edgewise decision yields a strong fixed-circuit certificate, while minimality profiles provide a complementary descriptive view of how edge effects are distributed across the selected circuit. Under selector-level cross-fit, edgewise minimality is interpreted foldwise or on a separately audited final circuit.

3.4 Power

Certificates also report the inferential resolution of the audit. For a target effect size δ , the framework computes minimum detectable effects under normal,

Hoeffding, and empirical-Bernstein rules, with the selected rule recorded in the certificate. The main experiments use the empirical-Bernstein rule, aligning the power calculation with the confidence intervals used for preservation and localization.

Interpretation. A certificate reports mechanistic evidence as a profile over preservation, localization, minimality, stability, robustness, power, and structural recovery. These dimensions can agree, separate, or conflict. The experiments show how profile-level reporting states mechanistic claims with explicit scope and uncertainty.

4 Experimental design

Implementation. The framework is instantiated in a consolidated artifact built on TransformerLens-based discovery integrations and protocol-aware certificate builders, with explicit protocol metadata, claim and certificate schemas, selector integrations, task loaders, and archived run packets.

Selectors and interventions. The main experiments use EAP-IG as the primary selector and EAP as a comparator on InterpBench, following current practice in scalable circuit localization [6, 12]. The primary intervention is counterfactual patching, with mean ablation and optimal ablation used as robustness follow-ups.

Tasks. We evaluate four representative settings. **IOI** on GPT-2 small is the flagship canonical case [15]. **ARC-Easy** on Gemma2 provides a realistic task with strong selector-level preservation. **Arithmetic subtraction** on Gemma2 provides a second non-IOI localization case. **InterpBench IOI** provides semi-synthetic transformers with ground-truth circuits, enabling direct comparison between behavioral support and structural recovery [5]. A final MCQA control is reported in the appendix.

Protocols. The main table reports true selector-level cross-fit, the primary selector-level protocol in the artifact. Selective-split comparisons are reported in the appendix and illustrate how the audited target affects the supported decision.

Thresholds and metrics. Each task uses pre-specified preservation and localization thresholds recorded in the audited run packets. The main certificates use the bounded backend score `prob_diff` in $[-1, 1]$ and the signed transformed audit loss `prob_diff_delta` in $[-2, 2]$. Preservation is assessed with one-sided upper confidence bounds, localization with one-sided lower confidence bounds, and power through empirical-Bernstein minimum detectable effects. Stability

Case	n_{audit}	$(\tau_{\text{pres}}, \tau_{\text{loc}})$	$[a, b]$	Interventions
IOI	1000	(0.3, 0.3)	$[-2, 2]$	$\{\text{cf}, \text{mean}\}$
ARC-Easy	570	(0.2, 0.2)	$[-2, 2]$	$\{\text{cf}, \text{mean}\}$
Arithmetic	137	(0.2, 0.2)	$[-2, 2]$	$\{\text{cf}, \text{mean}\}$
InterpBench	1000	(0.2, 0.2)	$[-2, 2]$	$\{\text{cf}, \text{mean}\}$

Table 1: Main selector-level audit design. The backend score is `prob_diff` in $[-1, 1]$, and the bounded audited loss is the signed transformed score `prob_diff_delta` in $[-2, 2]$. Splits are formed at the unit of independent task-instance rows.

Case	$ C_k $	med. [min,max]	$\hat{R}_{\text{pres}}/\text{UCB}$	Pres.	$\hat{R}_{\text{loc}}/\text{LCB}$	Loc.	Robust. (P/L)	Power	Jaccard
IOI	49	[48,52]	0.721 / 0.802	×	0.873 / 0.790	✓	× / ✓	✓	0.942
ARC-Easy	77	[70,84]	0.004 / 0.091	✓	0.004 / -0.082	×	✓ / ×	✓	0.821
Arithmetic	65	[50,72]	0.980 / 1.456	×	1.110 / 0.623	✓	× / ×	×	0.647
InterpBench / EAP-IG	59	[57,62]	0.161 / 0.251	×	0.440 / 0.325	✓	× / ✓	✓	0.791

Table 2: Main selector-level procedure audits. The size column reports the median foldwise selected-circuit size. Preservation and localization decisions are thresholded separately from the adequate-power flag, and robustness is decomposed into preservation robustness (P) and localization robustness (L) over the declared intervention family $\{\text{counterfactual}, \text{mean}\}$.

is summarized by mean pairwise Jaccard overlap and core/fringe decomposition. Splits are formed at the level of independent task-instance rows, before construction of derived patching evaluations.

5 Results

5.1 Selector-level evidence separates into regimes

Table 2 reports four selector-level regimes. For true selector-level cross-fit, the size column gives the median foldwise selected-circuit size rather than a single fixed-circuit size.

Localization-dominant regime. On GPT-2 small IOI, true selector-level cross-fit returns foldwise circuits with median size 49 and range $[48, 52]$, high selector stability (Jaccard 0.942), and adequate power (MDE 0.081). Localization is supported, with estimate 0.873 and one-sided lower bound 0.790, giving a margin of 0.490 above the threshold. Preservation falls short: the estimate is 0.721 and the one-sided upper bound is 0.802. Minimality fails under Holm-adjusted edgewise testing. The robustness vector follows the behavioral split: localization robustness passes over the declared intervention family $\{\text{counterfactual}, \text{mean}\}$, while preservation robustness fails. The resulting selector-level profile is localization without preservation. The same qualitative pattern appears on Gemma2 arithmetic subtraction, with weaker power.

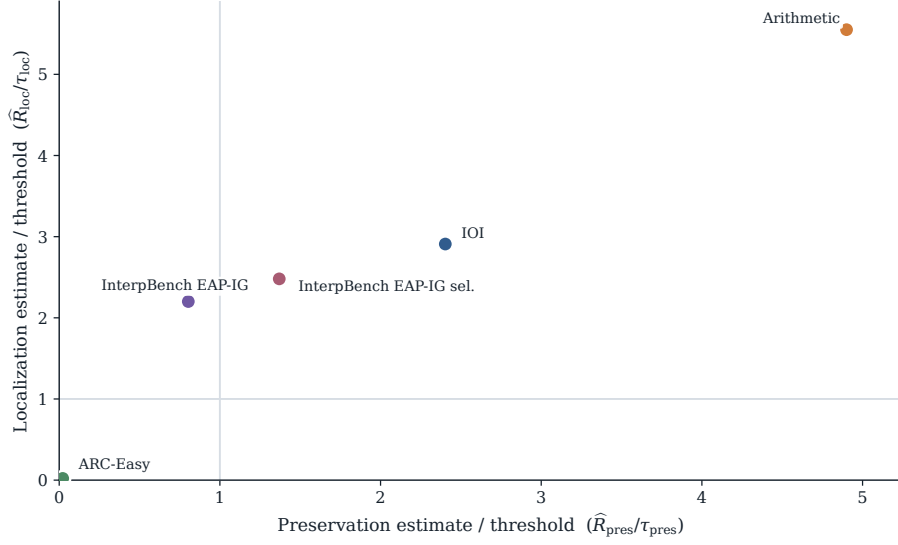


Figure 2: Preservation and localization across the main cases.

Preservation-dominant regime. On Gemma2 ARC-Easy, the selector-level audit exhibits the converse profile. The foldwise circuits have median size 77 and range [70, 84]. Preservation is supported, with estimate 0.004 and upper confidence bound 0.091, giving a margin of 0.109 below the threshold; preservation robustness also passes. Localization fails, with estimate 0.004 and a lower confidence bound below zero. The resulting profile is preservation without localization: the selected circuit preserves the task metric, while the audit does not concentrate the behavior in the selected subgraph.

Minimality and stability. All four main selector-level cases fail familywise minimality. Selector stability ranges from very high on IOI to moderate on arithmetic. Stability measures concentration of the selector distribution, while minimality measures edgewise essentiality under the audit. The main cases therefore include selectors that repeatedly return similar circuits whose audited edge effects still contain a broad unsupported fringe.

Power. The arithmetic case illustrates the role of power in the certificate. Localization passes under the chosen threshold, while the empirical-Bernstein MDE is much larger than in IOI or ARC-Easy. The certificate records the positive localization decision together with its lower inferential resolution.

Object	$ C $	Behav. (P/L)	Precision	F1
Ground truth	7	×/ ×	1.000	1.000
Full graph	1108	✓/ ×	0.006	0.013
Empty graph	0	×/ ×	0.000	0.000
EAP / true selector	38	×/ ✓	0.026	0.044
EAP-IG / true selector	53	×/ ✓	0.019	0.033

Table 3: InterpBench reports behavioral support alongside edge-level structural recovery. The selected EAP and EAP-IG circuits support localization under the audited intervention while remaining weak recoveries of the ground-truth edge set.

5.2 Behavioral support separates from structural recovery

InterpBench allows behavioral certificates to be compared against known ground-truth circuits. In this setting, the ground-truth circuit has perfect structural recovery by construction, whereas its behavioral audit depends on the audited intervention, score, thresholding convention, and graph granularity. The full graph preserves behavior but has low precision as a structural hypothesis. The selected EAP and EAP-IG circuits under true selector-level cross-fit localize behavior, although their structural precision and F1 remain low. Behavioral support and structural recovery are therefore reported as separate evidential dimensions.

The MI results should be read with this separation in view. A selector can identify a behaviorally meaningful subgraph with limited recovery of the true edge-level mechanism, and a structurally exact circuit can behave differently under the behavioral objective induced by a particular intervention design. When the benchmark ground-truth circuit fails the behavioral audit, the result indicates a mismatch between the structural target and the audited behavioral estimand. This mismatch is part of the evidence structure captured by the certificate.

5.3 Protocol choice changes the supported claim

The appendix compares selective split with true selector-level cross-fit on shared tasks. On ARC-Easy, the selector-level protocol supports preservation where the selected-circuit audit did not; on both InterpBench methods, it supports localization where the selected-circuit audit did not; and on IOI, both protocols yield localization without preservation. These comparisons place the audit protocol inside the scientific claim, since changing the protocol changes the audited target.

5.4 Minimality profiles reveal descriptive high-effect tails

Although the Holm-adjusted familywise minimality decision fails on all main selector-level circuits, the minimality profiles reveal heavy-tailed edge-effect

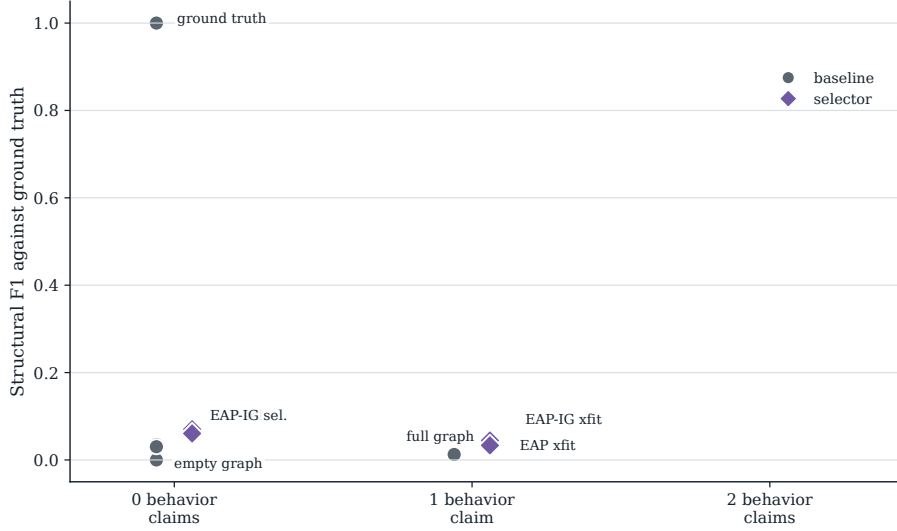


Figure 3: Behavioral support and structural recovery on InterpBench.

distributions. For IOI, the median edge effect is positive and the upper quartile is substantially larger than zero; the same qualitative shape appears on arithmetic and InterpBench. The appendix visualizes these profiles and the resulting picture is a descriptive high-effect tail embedded in a broad low-effect fringe, a structure that a single pass/fail minimality decision cannot capture.

6 Conclusion

Circuit discovery produces adaptive objects: subgraphs selected through task definitions, counterfactual constructions, scores, selectors, thresholds, and graph abstractions. The corresponding scientific unit of evidence is the mechanistic claim, which couples the selected circuit to the model, task, intervention design, audit protocol, audited property, and selection procedure under which it was obtained. A Circuit Certificate records that claim together with its uncertainty calculation, stability summary, minimality analysis, robustness vector, power calculation, and structural-recovery results when available.

The theory supplies the inferential discipline for this claim-level view. Selective split yields finite-sample one-sided certification for any measurable selector under independent audit data. True selector-level cross-fit gives valid foldwise out-of-fold audits for circuits selected without the audited fold and targets the behavior of the selection procedure across repeated rediscovery. Edgewise minimality can be audited with familywise error control, and power is reported in the same finite-sample regime as the behavioral bounds. Together, these components define circuit evaluation as a protocol-indexed statistical audit.

The experiments separate the evidential dimensions captured by the certificates. Preservation, localization, minimality, stability, robustness, power, and structural recovery vary across canonical and benchmarked settings. GPT-2 small IOI yields strong selector-level localization with high stability but unsupported preservation and minimality. Gemma2 ARC-Easy yields preservation without localization. InterpBench shows behavioral support coexisting with weak edge-level structural recovery. The certificate format reports these regimes as distinct properties of the claim, instead of compressing them into a single circuit-quality score.

Circuit Certificates provide an executable reporting standard for mechanistic interpretability by identifying the claim supported by a discovered circuit, attaching that claim to the protocol that produced the evidence, and reporting uncertainty in the vocabulary of the mechanistic property under audit. The resulting analysis reports auditable mechanistic evidence together with the selected subgraph.

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A Proofs and formal details

A.1 Set-up

Let $Z = (X, \tilde{X}, Y)$ denote a generic task tuple, and let $w(c, Z)$ be a bounded measurable loss associated with a circuit c and a sampled tuple Z . Preservation and localization losses are special cases of this template. For a selected circuit C , the target risk is

$$R(C) = \mathbb{E}_{Z \sim \mathcal{D}}[w(C, Z)].$$

Under selective split, the selected circuit depends only on selector-visible data and is independent of the audit sample. Under true selector-level cross-fit, each foldwise selected circuit is independent of its own audit fold by construction.

A.2 Proof of Theorem 2

Proof. Condition on the selector-visible data S and therefore on the selected circuit $C = g(S)$. Since the audit sample is independent of S , the audit observations are i.i.d. from $\mathcal{D}$ conditional on S . The variables $W_i = w(C, Z_i)$ are therefore i.i.d. and bounded in $[a, b]$ conditional on S , with mean $R(C)$. Any valid one-sided confidence procedure for bounded i.i.d. means applies conditionally on S , including the empirical-Bernstein bound used in the artifact. Integrating over S gives the marginal coverage statement. $\square$

A.3 Proof of Proposition 3

Proof. Fix the fold partition and a fold k . The circuit C_k is selected without using the data in $\mathcal{I}_k$. Hence, for every $i \in \mathcal{I}_k$, the held-out tuple Z_i is independent

of the circuit used to audit it. Conditional on C_k , the held-out losses $\{w(C_k, Z_i) : i \in \mathcal{I}_k\}$ are i.i.d. bounded variables with conditional mean

$$R_k = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \mathbb{E}[w(C_k, Z_i) \mid C_k].$$

Therefore any valid one-sided confidence procedure for bounded i.i.d. means applies on the held-out fold, including the empirical-Bernstein rule used in the artifact. $\square$

Remark on aggregation. The aggregated selector-level estimator remains a useful procedure-level summary of foldwise out-of-fold losses. However, because an observation in fold j may be used to select the circuit audited on another fold $k \neq j$, conditioning on all selected circuits does not in general restore independence across all aggregated terms for arbitrary adaptive selectors. We therefore report the aggregate as a procedure-level estimator without claiming an unconditional finite-sample concentration theorem in this setting.

A.4 Proof of Proposition 4

Proof. Condition on the selected circuit and on the data used to select it, leaving the edgewise audit data random. By assumption, each edgewise p -value is valid for its corresponding null hypothesis. Holm’s step-down procedure controls the familywise error rate at level α for any finite family of valid p -values. The conditional familywise guarantee therefore holds, and marginal control follows by averaging over the selection randomness. $\square$

A.5 Minimality profiles are monotone

Proposition 5. *For any fixed circuit C , the function $\epsilon \mapsto \text{MinProf}_C(\epsilon)$ is monotone non-increasing.*

Proof. If $\epsilon_1 \leq \epsilon_2$, then for every edge $e \in C$,

$$\mathbf{1}[\Delta_e(C; \eta) > \epsilon_2] \leq \mathbf{1}[\Delta_e(C; \eta) > \epsilon_1].$$

Summing over edges and dividing by $|C|$ gives the result. $\square$

A.6 Preservation and localization are logically distinct

Proposition 6. *There exist tasks, intervention designs, and circuits for which preservation holds and localization fails, and there exist tasks, intervention designs, and circuits for which localization holds and preservation fails.*

Proof. Normalize the clean-model score to 1. Consider a circuit whose selected-subgraph score is 0.95 and whose complement score is 0.94. The preservation discrepancy is 0.05, and the localization discrepancy is 0.06. With thresholds (0.1, 0.1), preservation holds and localization fails.

Conversely, consider a circuit whose selected-subgraph score is 0.05 and whose complement score is 0. The preservation discrepancy is 0.95, and the localization discrepancy is 1.0. With the same thresholds, localization holds and preservation fails. $\square$

A.7 Minimum detectable effects

Let n be the audit sample size, let $\hat{\sigma}^2$ be the audit variance estimate, and let $[a, b]$ be the range of the bounded loss. The artifact reports three minimum detectable effects:

$$\text{MDE}_{\text{normal}} = \frac{(z_{1-\alpha} + z_{\text{power}})\hat{\sigma}}{\sqrt{n}}, \quad \text{MDE}_{\text{Hoeffding}} = (b - a)\sqrt{\frac{\log(1/\alpha)}{2n}},$$

$$\text{MDE}_{\text{EB}} = \sqrt{\frac{2\hat{\sigma}^2 \log(3/\alpha)}{n}} + \frac{3(b - a) \log(3/\alpha)}{n}.$$

The main experiments use MDE_{EB} , aligning the power rule with the one-sided confidence bounds used for preservation and localization.

B Additional results

B.1 Protocol comparisons

Table 4 compares the audited protocols on shared tasks.

Case	Left protocol	Right protocol	Pres. (L→R)	Loc. (L→R)	$\Delta \hat{R}_{\text{pres}}$	$\Delta \hat{R}_{\text{loc}}$
IOI	selective split	true selector	×→×	✓→✓	0.167	0.022
InterpBench / EAP-IG	selective split	true selector	×→×	×→✓	-0.113	-0.056
InterpBench / EAP	selective split	true selector	×→×	×→✓	-0.077	-0.066
ARC-Easy	selective split	true selector	×→✓	×→×	-0.000	-0.000
Arithmetic	selective split	true selector	×→×	×→✓	0.062	0.085

Table 4: Protocol comparisons on shared tasks. The left-hand protocol audits one selected circuit, while the right-hand protocol audits the selector as a procedure. The transitions therefore reflect a change in audited target, not one protocol dominating a fixed-circuit claim.

B.2 Selector-level detailed values

Table 5 reports the selector-level values used for the threshold decisions.

B.3 InterpBench full suite

Table 6 gives the full InterpBench ground-truth suite.

Case	95% UCB_{pres}	m_{pres}	95% LCB_{loc}	m_{loc}	MDE	Jaccard
IOI / true selector	0.802	-0.502	0.790	0.490	0.081	0.942
InterpBench / EAP-IG / true selector	0.251	-0.051	0.325	0.125	0.091	0.791
ARC-Easy / true selector	0.091	0.109	-0.082	-0.282	0.087	0.821
Arithmetic subtraction / true selector	1.456	-1.256	0.623	0.423	0.476	0.647

Table 5: Detailed selector-level values. The margins are $m_{pres} = \tau_{pres} - UCB_{pres}$ and $m_{loc} = LCB_{loc} - \tau_{loc}$, so positive values indicate threshold clearance.

Object	Protocol	$ C $	Pres.	Loc.	Precision	Recall	F1
empty graph	fixed_holdout	0	×	×	0.000	0.000	0.000
full graph	fixed_holdout	1108	✓	×	0.006	1.000	0.013
ground truth	fixed_holdout	7	×	×	1.000	1.000	1.000
random same size eap seed0	fixed_holdout	50	×	×	0.020	0.143	0.035
random same size eap seed1	fixed_holdout	50	×	×	0.000	0.000	0.000
random same size eap seed2	fixed_holdout	50	×	×	0.000	0.000	0.000
random same size eapig seed0	fixed_holdout	59	×	×	0.017	0.143	0.030
random same size eapig seed1	fixed_holdout	59	×	×	0.000	0.000	0.000
random same size eapig seed2	fixed_holdout	59	×	×	0.000	0.000	0.000
eap selective split	selective_split	50	×	×	0.040	0.286	0.070
eap true crossfit selector	true_crossfit_selector	38	×	✓	0.026	0.143	0.044
eapig selective split	selective_split	59	×	×	0.034	0.286	0.061
eapig true crossfit selector	true_crossfit_selector	53	×	✓	0.019	0.143	0.033

Table 6: InterpBench ground-truth suite. The suite contrasts the ground-truth circuit, full and empty graphs, graph-size-matched random circuits, and circuits discovered by EAP and EAP-IG under selective-split and selector-level protocols.

B.4 Minimality profiles

Table 7 reports empirical minimality quantiles for the main cases.

Case	$q_{0.25}$	$q_{0.50}$	$q_{0.75}$	$q_{1.00}$
IOI / true selector	-0.0078	0.0029	0.0205	0.1189
ARC-Easy / true selector	-0.0000	-0.0000	0.0000	0.0010
Arithmetic / true selector	-0.0005	0.0014	0.0086	0.1758
InterpBench / EAP-IG / true selector	-0.0037	0.0106	0.0389	0.1861

Table 7: Empirical quantiles of edgewise minimality effects. The profiles describe a high-effect tail together with a wide low-effect bulk.

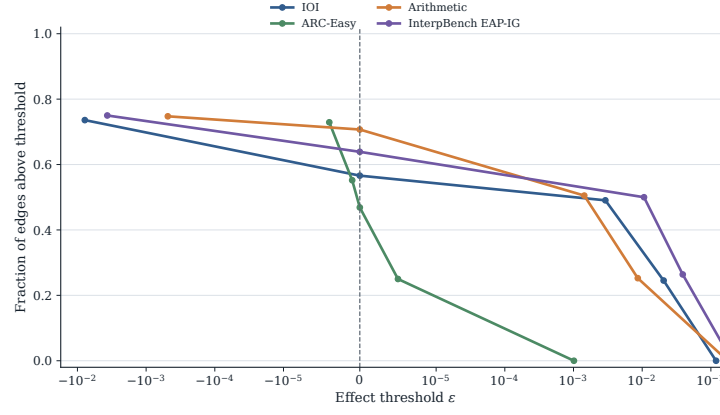


Figure 4: Minimality profiles for the main cases.

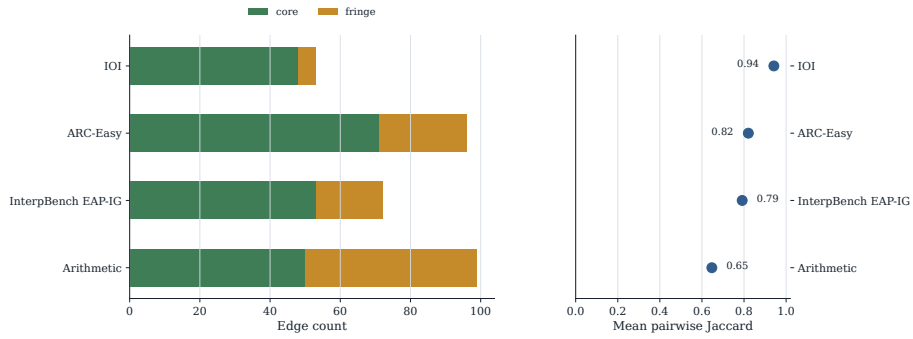


Figure 5: Selector stability for the main cases.

	Pres	Loc	Min	Stab	Loc-R	Pres-R	Pow	F1
GPT-2 IOI true cross-fit	FAIL	PASS	FAIL	0.94	PASS	FAIL	PASS	n/a
InterpBench IOI EAP-IG true cross-fit	FAIL	PASS	FAIL	0.79	PASS	FAIL	PASS	0.033
InterpBench IOI EAP true cross-fit	FAIL	PASS	FAIL	0.56	PASS	FAIL	PASS	0.044
Gemma2 ARC-Easy true cross-fit	PASS	FAIL	FAIL	0.82	FAIL	PASS	PASS	n/a
Gemma2 Arithmetic true cross-fit	FAIL	PASS	FAIL	0.65	FAIL	FAIL	FAIL	n/a

Figure 6: Claim matrix for the main cases.

C Artifact, provenance, and reproducibility

The submission is supported by a consolidated artifact containing the code and archived run packets used for the reported experiments.

Protocol metadata. Each certificate records the audit protocol, selector-independence status, and identifiers for the discovery, calibration, and audit splits used in the run. These fields distinguish fixed-circuit evidence from selector-level evidence throughout the archive.

Execution. The artifact README provides installation, a validation command, a minimal end-to-end run, and direct paths to the archived run packets used by the reported cases.